torch.ormqr#

https://pytorch.org/docs/stable/generated/torch.ormqr.html#torch.ormqr

Description:

Computes the matrix-matrix multiplication of a product of Householder matrices with a general matrix. Multiplies a m×nm \times nm×n matrix C (given by other) with a matrix Q, where Q is represented using Householder reflectors (input, tau). See Representation of Orthogonal or Unitary Matrices for further details. If left is True then op(Q) times C is computed, otherwise the result is C times op(Q). When left is True, the implicit matrix Q has size m×mm \times mm×m. It has size n×nn \times nn×n otherwise. If transpose is True then op is the conjugate transpose operation, otherwise it’s a no-op. Supports inputs of float, double, cfloat and cdouble dtypes. Also supports batched inputs, and, if the input is batched, the output is batched with the same dimensions.

Function Signature

```
torch.ormqr(input, tau, other, left=True, transpose=False,
*, out=None) → Tensor#

Parameters

Keyword Arguments
```

Parameters

Keyword

out (Tensor, optional) – the output Tensor. Ignored if None. Default: None.

Notes & Warnings

> **NOTE** See also torch.geqrf() can be used to form the Householder representation (input, tau) of matrix Q from the QR decomposition.

> **NOTE** Note This function supports backward but it is only fast when (input, tau) do not require gradients and/or tau.size(-1) is very small. ``

Detailed Documentation

Documentation

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Multiplies a m×nm \times nm×n matrix C (given by other) with a matrix Q, where Q is represented using Householder reflectors (input, tau). See Representation of Orthogonal or Unitary Matrices for further details.

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Supports inputs of float, double, cfloat and cdouble dtypes. Also supports batched inputs, and, if the input is batched, the output is batched with the same dimensions.

torch.geqrf() can be used to form the Householder representation (input, tau) of matrix Q from the QR decomposition.

This function supports backward but it is only fast when (input, tau) do not require gradients and/or tau.size(-1) is very small. ``

input (Tensor) – tensor of shape (*, mn, k) where * is zero or more batch dimensions and mn equals to m or n depending on the left.

tau (Tensor) – tensor of shape (*, min(mn, k)) where * is zero or more batch dimensions.

other (Tensor) – tensor of shape (*, m, n) where * is zero or more batch dimensions.

left (bool) – controls the order of multiplication.

transpose (bool) – controls whether the matrix Q is conjugate transposed or not.

out (Tensor, optional) – the output Tensor. Ignored if None. Default: None.
